

Imputation Matters: Evaluating the Impact of Missing Data Strategies on Interpretability in Clinical Time Series Models

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Abstract

In healthcare time series forecasting, missing data is a prevalent challenge that undermines both model performance and interpretability. This paper investigates how different imputation strategies affect not only the predictive accuracy of machine learning models but also the internal logic behind their decisions. We propose a systematic evaluation using LSTM-based models trained on the PhysioNet Sepsis Challenge dataset, applying three common imputation methods, mean imputation, backward/forward fill, and linear interpolation, across varying feature sets. To assess interpretability, we employ Layer-wise Relevance Propagation (LRP), revealing that the choice of imputation significantly alters feature relevance patterns, even when predictive metrics remain similar. Our findings show that imputation is not a neutral preprocessing step; it fundamentally shapes what models learn and how they explain their outputs. These insights highlight the need for careful imputation design in clinical AI systems, especially where transparency and trust are critical.

CCS Concepts

• **Computing methodologies** → **Neural networks; Supervised learning by classification**; • **Mathematics of computing** → **Time series analysis**; • **Human-centered computing** → **Heat maps**.

Keywords

clinical time series; machine learning; explainability; missing data; imputation; LRP; interpretability; robustness

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1 Introduction

In real-time systems such as those used in healthcare, data is often collected through sensors or manual documentation. Increasingly,

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these systems are enhanced with machine learning models to support clinical decision-making. However, missing data is a prevalent challenge in this domain. In clinical settings, data may be missing for various reasons: sensor failures, limited resources for patient monitoring, human error, or irregular sampling rates inherent to time series data [2].

In addition to missing data, another critical concern is the lack of transparency in machine learning models. In high-stakes domains like healthcare, the “black-box” nature of AI hinders trust among practitioners, especially when these models influence clinical decisions. Explainability becomes vital not only to promote trust but also to ensure that models make decisions for clinically meaningful reasons [9].

These two challenges, missing data and explainability, are deeply intertwined. Imputing missing values introduces synthetic information into the data stream, which can alter the behavior of the model during training and inference. Consequently, the choice of imputation method may introduce bias into the model. In this work, we expect that different imputation strategies will not only affect model performance, but also influence how models interpret and explain clinical data.

This paper investigates how different imputation strategies impact both model performance and interpretability. We train multiple LSTM-based models on various versions of the PhysioNet Sepsis Challenge dataset [12], each filtered by missing data and processed with a different imputation method. By applying Layer-wise Relevance Propagation (LRP), we analyze how the relevance scores assigned to features and time steps vary across imputations. Our results show that imputations can not only affect predictive performance but also lead to different feature importance patterns, even for the same patient. This has implications not only for the reliability and fairness of clinical AI models, but more generally for the choice of imputation methods in any ubiquitous system.

2 Related Work

Clinical datasets are often time series in nature, capturing a patient’s health development over time. Missing data is a common issue, and in many cases, it is not missing at random. For instance, certain laboratory tests may only be performed when a patient’s condition worsens, resulting in informative missingness. Traditional imputation strategies frequently overlook this nuance, which can lead to unintended biases in downstream analyses [2].

2.1 Strategies for Handling Missing Data

There are three widely used strategies to address missing data: (1) *Feature removal*, which excludes features that exhibit a high proportion of missing values; (2) *Sample removal*, which discards

records containing missing entries; and (3) *Imputation*, where missing values are estimated and filled in based on statistical or heuristic methods.

Feature and sample removal rely solely on observed data, avoiding the risk of introducing artificial values, but can lead to information loss, especially when rare but informative patterns are discarded. In contrast, imputation retains all cases but may introduce bias or artifacts [6].

In time series analysis, commonly used imputation methods include the following: *mean imputation*, which replaces missing values with the global mean of the corresponding feature; *forward/backward fill*, which propagates the most recent past or future value to fill gaps; and *linear interpolation*, which estimates missing values based on a linear trend between adjacent observed values [1].

2.2 Explainability Mechanisms and Imputation

Explainability methods in machine learning can generally be divided into intrinsic and post hoc techniques [8]. Intrinsic methods incorporate interpretability directly into the model architecture or learning objective, for example through constraints or multi-task learning. Post hoc techniques, by contrast, analyze already trained models to provide insight into their decision-making.

These methods include perturbation-based approaches such as SHAP (SHapley Additive exPlanations), which systematically alter input features and measure changes in the output [7], and gradient-based approaches like LRP, which backpropagate the model's prediction to assign relevance scores to input features [8].

In the case of time series, explainability becomes more challenging due to the sequential dependencies and the interaction between features across time. LRP, however, explains a prediction by calculating the relevance of each input value by passing backwards through the model. This allows for fine-grained insight into each input timestamp and feature.

Analyzing explanations of AI and establishing trust is a difficult task especially in health care [13].

In this paper, explainability methods are not used to determine whether the model learns the correct attributes of the domain, but rather to gain insight into how imputation and feature removal influence what the model learns.

2.3 PhysioNet Sepsis Dataset

Sepsis is a deadly disease that is difficult and costly to detect early. Early detection, however, is crucial for successful treatment. For the PhysioNet/Computing in Cardiology Challenge 2019, a dataset was released along with a challenge to predict sepsis up to six hours before onset. The dataset consists of publicly available data from two hospitals, covering over 40,000 patients and including 40 features such as vital signs, laboratory values, and demographic information. The task is a binary classification: predicting whether a patient will develop sepsis within a six-hour time window [5, 11, 12].

The dataset contains many irregularly measured features, as it is based on real hospital data. To evaluate the winners of the challenge, a specific scoring metric was introduced [10]. This paper does not introduce a new baseline for solving the dataset. Instead, the dataset is used to evaluate, on real-world data, how imputation and data omission influence the predictions of machine learning

models and more importantly, to analyze which features are learned to be important by these models.

3 System

The proposed system follows a standard machine learning pipeline, adapted to address the challenges of missing data in clinical time series. The full pipeline consists of four main stages: data processing, imputation and scaling, model training, and evaluation with interpretability.

3.1 Data Processing

The dataset is divided into training, validation, and test splits in a 70%-20%-10% ratio. To avoid data leakage, patient-level stratification is enforced, ensuring that no single patient appears in more than one split.

We first analyze the training split to identify features with a high proportion of missingness. A threshold-based filtering is applied: if a feature exceeds a predefined percentage of missing values, it is removed entirely from all splits. This allows us to evaluate models across different feature configurations (e.g., 40, 27, 14, and 10 retained features).

3.2 Imputation and Scaling

Three baseline imputation strategies are applied independently to the datasets:

Mean imputation replaces missing values with the global mean of each feature, calculated from the training set.

Forward/backward fill (*Bfill/fill*) replaces each missing value using the nearest previous or next available value in the time dimension.

Linear interpolation (*Linear*) is applied by first interpolate linearly and applying forward and backward fill to fill any missing gaps, providing smooth transitions between real observations.

After imputation, all features are standardized using z-score normalization, where the mean and standard deviation are computed from the training set.

3.3 Model Architecture and Training

The model used is a Long Short-Term Memory (LSTM) neural network tailored for binary classification of sepsis. The architecture includes an LSTM layer for processing the input time series, with configurable hidden size and number of layers, followed by dropout for regularization. Temporal pooling is applied across the sequence length to obtain a fixed-size representation, which is then passed to a fully connected output layer with a single neuron producing the logit for binary classification.

To address class imbalance, since sepsis is a minority class, focal loss is used to better learn the hard-to-classify cases. Additionally, threshold tuning on the validation set is employed to optimize the F1 score.

The model is trained using mini-batch gradient descent with the Adam optimizer, and early stopping is applied based on validation loss. Each configuration (imputation method combined with feature set) is trained independently to ensure comparability, resulting in 12 different datasets and models, as we explore four different feature sets and three different imputation strategies.

3.4 Evaluation and Explainability

After training, each model is evaluated on the held-out test set. The reported metrics include Area Under the Receiver Operating Characteristic Curve (AUROC), accuracy, F1 score, precision, and recall. In addition, we apply LRP to interpret the predictions. LRP generates relevance heatmaps for each input feature across all time steps, helping to analyze how different imputation methods affect model attention and decision logic.

4 Results

In this section, we present the effects of different imputation strategies and feature set sizes on the performance of the LSTM-based sepsis prediction models. Each configuration was evaluated using AUROC, F1 score, precision, and recall. A summary of the best-performing models for each imputation strategy and feature count combination is shown in Table 1.

Table 1: Model performance metrics (AUROC, Accuracy, Precision, Recall, F1 Score) for feature counts 10, 27, and 40.

Model	AUROC	Accu.	Precis.	Recall	F1 Score
Bfillfill_10	0.6653	0.9177	0.0633	0.1707	0.0924
Linear_10	0.6611	0.8624	0.0526	0.2707	0.0880
Mean_10	0.6800	0.8531	0.0553	0.3102	0.0939
Bfillfill_27	0.7271	0.9364	0.0964	0.1901	0.1280
Linear_27	0.7172	0.9522	0.1102	0.1342	0.1210
Mean_27	0.7333	0.9244	0.0844	0.2113	0.1207
Bfillfill_40	0.6881	0.9617	0.1306	0.0995	0.1129
Linear_40	0.6968	0.9003	0.0627	0.2195	0.0975
Mean_40	0.7453	0.9172	0.0945	0.2766	0.1409

All metrics were calculated on the test data split, which was excluded during training. This test set consists of 4,034 patients, of which only 7.6% are sepsis cases. Out of 156,830 total timestamps, only 2,730 contain the sepsis label.

This class imbalance explains the relatively low values for accuracy, precision, recall, and F1 score. Accuracy is high primarily because non-sepsis cases are easier for the model to predict.

The number of features directly influences the F1 score. Using 27 features leads, on average, to the best performance, while using only 10 features results in the worst. The only exception is the Mean_40 model, which performs strongly across F1 score, AUROC, and recall.

This evaluation shows that using many features with artificially imputed values can deteriorate model performance. However, it also demonstrates that certain imputations, such as in the Mean_40 model, can outperform others in specific metrics. In this case, having consistent, imputed values across sparse features may have helped reduce noise and improve inference.

As part of the training process, a threshold-based approach was applied to optimize the F1 score and AUROC by learning the classification threshold on the validation split. However, this strategy may not be ideal in clinical settings, where higher recall is typically preferred to minimize false negatives. In such contexts, the training

objective could be adjusted to prioritize recall, even at the cost of overall F1 performance.

Overall, the results confirm the initial assumption: both imputation method and feature count have a non-trivial influence on model performance. Each imputation strategy responds differently to changes in feature availability.

4.1 Qualitative Analysis of LRP Explanations

To assess the impact of imputation on model interpretability, we generated LRP heatmaps for a representative test sequence from a patient across nine different model configurations. The sequence shown in Figure 2 was selected because each model produced comparable outputs for this case.

The qualitative analysis shows that depending on the imputation method and the number of features, models learn different aspects of the dataset. The Mean_40 model applies low but consistent relevance across nearly all data points. However, one clear outlier is the FiO₂ (fraction of inspired oxygen), which represents the amount of oxygen a patient inhales. It is also noticeable that individual timestamps can carry significant relevance, such as the temperature reading at timestamp nine.

The Linear_40 model shows almost the opposite behavior of the Mean_40 model. This suggests that, regardless of imputation method, FiO₂ is learned as an important feature for sepsis classification, but it may contribute either positively or negatively to the prediction. The Linear_40 model assigns near-zero relevance to most features, indicating that its decision is based almost entirely on FiO₂ for this particular sequence. In contrast, the Bfillfill_40 model assigns most relevance to timestamps four through six, without any single feature standing out.

These differences already indicate that the imputation method directly affects how models interpret inputs, and that models can arrive at similar outputs with very different internal explanations. For the 27-feature case, the relevance patterns in Linear and Mean models remain similar, but their outputs no longer reach the learned classification threshold. In Mean_27, timestamps four and five contain several highly relevant values.

In the reduced 10-feature datasets, all three models correctly predict sepsis, and all assign high relevance to the patient’s age. Interestingly, both Bfillfill and Mean also identify systolic blood pressure (SBP) as an important feature.

Overall, the qualitative evaluation shows that both feature count and imputation strategy can lead to completely different patterns of what the model learns during training. At the same time, it also reveals that certain patterns, such as relevance at specific timestamps or consistent focus on certain features, can emerge across different model configurations.

4.2 Quantitative Analysis of Relevance Shifts

Beyond individual examples, we conducted a quantitative comparison of how relevance distributions shift across imputation methods and feature subsets. For each model, we aggregated LRP relevance scores across the entire test set and computed the average relevance assigned to each feature, focusing specifically on sepsis cases. We then analyzed the variance in feature relevance rankings across different imputations to assess explanation consistency.

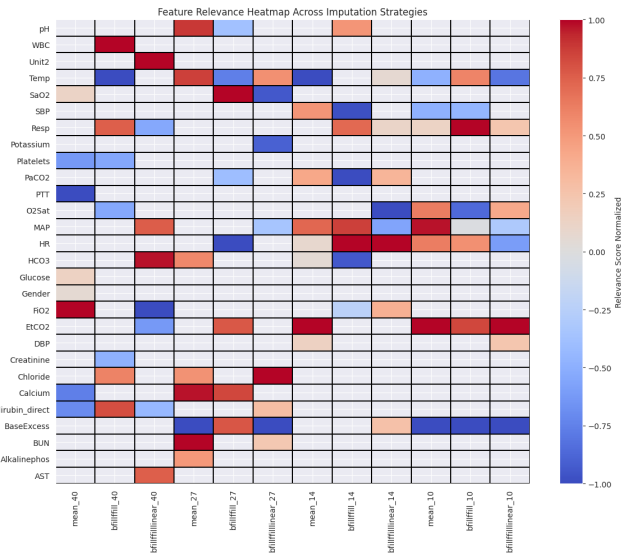


Figure 1: Heatmap of the top and bottom 4 feature relevance scores across imputation strategies. Positive values indicate strong relevance toward a sepsis prediction, while negative values indicate strong relevance against it.

Our findings reveal that, out of all 39 features, only 28 appear as either top-4 or bottom-4 features in any configuration. As shown in Figure 1, some features, such as temperature and mean arterial pressure, are consistently highlighted across multiple models. However, the overall distribution of relevance varies noticeably between imputations.

When calculating the percentage contribution of normalized positive relevance, we found that in the `Linear_14` model, temperature which is a vital sign, accounts for 21% of total relevance across all test data. In contrast, the `Mean_27` model highlights three laboratory values and temperature, each contributing approximately 5% on average.

The feature `FiO2`, which was identified as relevant in the qualitative evaluation, is also confirmed quantitatively as one of the most important indicators for both sepsis and non-sepsis predictions across these specific models.

These results highlight an important implication for clinical deployment: explanation-based trust in AI systems may be significantly influenced by the choice of preprocessing, even when performance metrics remain similar. For reliable interpretability, further research is needed to understand how these most relevant features align with clinical knowledge in sepsis diagnostics.

5 Limitations and Future Work

This study deliberately used a fixed LSTM architecture and training configuration across all experiments to isolate the influence of imputation strategies. While this ensures comparability, future work could explore how architectural or hyperparameter variations interact with interpretability outcomes.

We focused on three simple and widely-used imputation methods, mean imputation, forward/backward fill, and linear interpolation. While these represent common practice in preprocessing pipelines, more advanced approaches such as Focalize K-NN [1], GRU-D [4], or BRITS [3] could provide additional insights and should be considered in follow-up studies.

Furthermore, our analysis did not explicitly attribute the underlying missingness mechanisms [6], nor did it quantify potential statistical biases introduced by the imputation methods (e.g., variance shrinkage through mean imputation). Both aspects deserve further investigation to better understand how imputation affects data distributions and, ultimately, model behavior.

Finally, the interpretability analysis was conducted without formal input from clinical domain experts. Integrating clinician feedback could help assess the medical plausibility of relevance patterns and their trustworthiness in real-world decision support. Applying the framework to diverse clinical datasets could further validate the generalizability of our findings.

6 Conclusion

This paper investigated the impact of imputation strategies and feature availability on both the predictive performance and interpretability of LSTM-based models for early sepsis detection in clinical time series data. Through a systematic evaluation of three imputation techniques, mean imputation, forward/backward fill, and linear interpolation, we demonstrated that preprocessing choices have measurable effects not only on accuracy and F1 scores, but also on the internal decision logic of the model, as revealed by LRP explanations.

Our qualitative analysis of LRP heatmaps showed that the same patient can receive drastically different feature relevance attributions depending on the imputation method used. While some patterns were recognizable across imputations and feature sets, the relevance distributions varied significantly. The quantitative analysis further confirmed that relevance stability differs across imputation strategies, supporting the insights from the qualitative evaluation.

These findings emphasize the importance of carefully selecting imputation methods in clinical machine learning pipelines. Imputation is not a neutral preprocessing step, as it can significantly influence how models learn and explain their predictions. In sensitive domains like healthcare, this has important implications for transparency, reproducibility, and ultimately, patient safety.

Future work could explore more advanced or domain-specific imputation methods, such as generative approaches or learned imputers, and extend this analysis to other explainability techniques beyond LRP. Additionally, incorporating uncertainty quantification or explanation consistency metrics could help practitioners better assess model trustworthiness in the presence of missing data.

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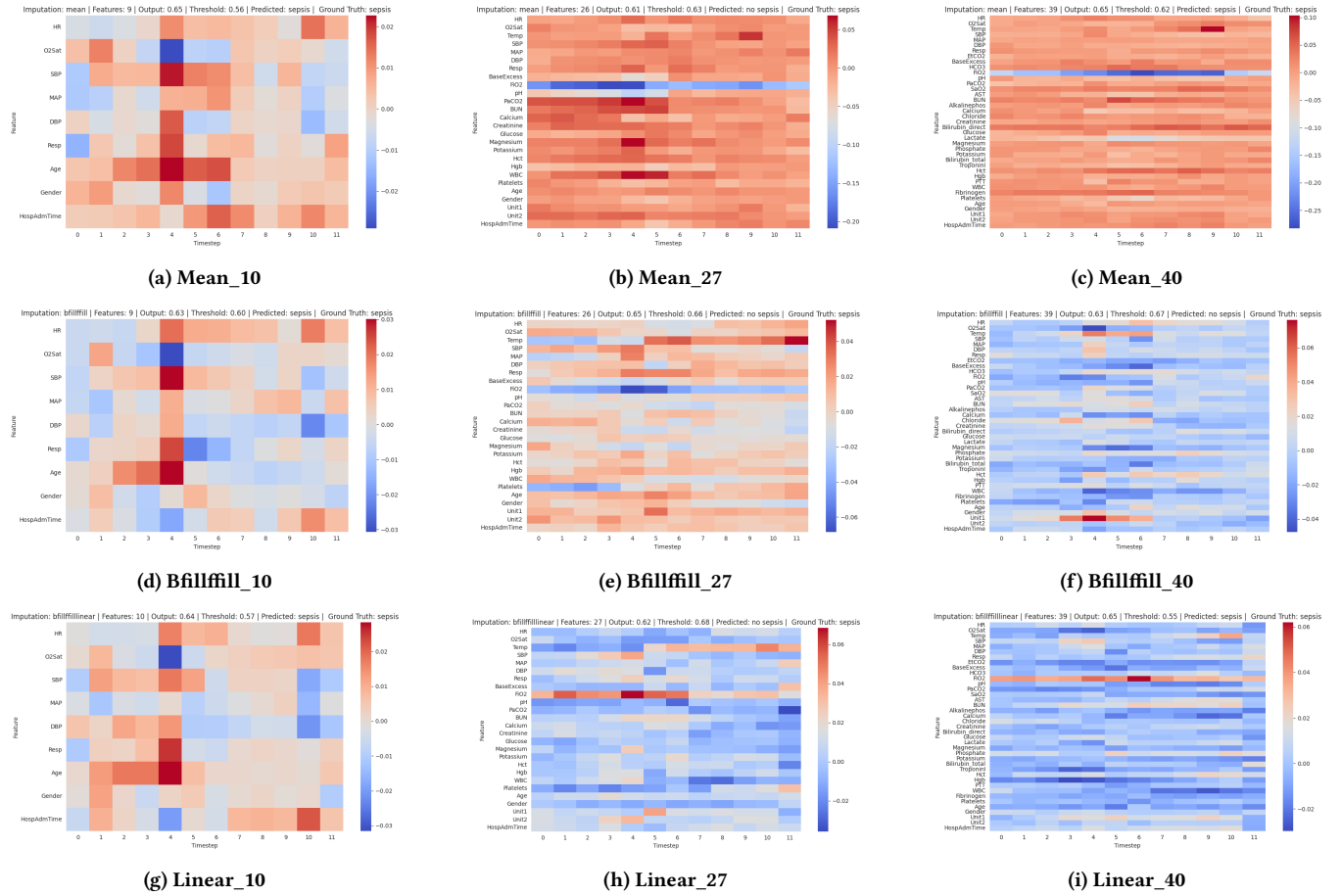


Figure 2: Relevance heatmaps using LRP for a single patient across all 9 model configurations (3 imputation strategies $\times$ 3 feature sets). Each feature map corresponds to the same patient instance in the respective dataset and shows relevance with respect to the sepsis class. Red values (positive relevance) indicate features and timestamps that contribute to a sepsis prediction, while blue values (negative relevance) indicate features and timestamps that support a non-sepsis prediction. The threshold line represents the classification threshold learned during validation, while the output is the model's predicted value.

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